

Qualifying Exam in Microeconomics
Arizona State University
June 2021

There are four questions and you have four hours to complete the exam. If you believe that a question is in error, please explain the error and continue working.

1. Consider an exchange economy with two goods. Its aggregate excess demand function is given by $z : \mathbb{R}_{++}^2 \rightarrow \mathbb{R}^2$ where $z(p_1, p_2) = (-1, \frac{p_1}{p_2})$ for all $(p_1, p_2) \gg 0$.
 - (a) State whether z satisfies each of the following properties. Either provide a proof or a counterexample.
 - i. Continuity.
 - ii. Homogeneity of degree zero.
 - iii. Walras' Law.
 - iv. Bounded below.
 - v. Boundary condition.
 - (b) Does this economy have a Walrasian equilibrium (i.e. $\exists (p_1^*, p_2^*) \gg 0$ such that $z(p_1^*, p_2^*) = 0$)? Either argue that it does (by proving it or appealing to the appropriate theorem) or demonstrate that it does not.

2. A person must take an action $a \in \{a_1, a_2\}$ at date 1 and an action $b \in C(a) \subseteq \{b_1, b_2\}$ at date 2, where for $a \in \{a_1, a_2\}$, $C(a)$ is the set of actions in $\{b_1, b_2\}$ that are available at date 2. Action a_1 is a *flexible* action: $C(a_1) = \{b_1, b_2\}$; that is, if a_1 is chosen at date 1, then both b_1 and b_2 are available at date 2. Action a_2 is an *inflexible* action: $C(a_2) = \{b_2\}$; that is, if a_2 is chosen at date 1, then only b_2 is available at date 2.

There are two states of the world, s_1 and s_2 . The person's prior belief that the state is s_2 is $1/2$, and the person's vN-M utility is

$$u(a, b, s) = u_1(a, s) + u_2(b, s), \quad (1)$$

where u_t is date- t utility (including any discounting). Impose these assumptions on the person's preferences.

- At date 2, if the state is known, the person wants to match the action with the state: if the state is s_i , then the person strictly prefers b_i to b_j , for $j \neq i$, $i = 1, 2$.
- In addition, the person strictly prefers action b_2 to b_1 at date 2 at the prior belief of $1/2$.

Between dates 1 and 2, the person observes a signal $y \in \{y_0, y_1, y_2\}$ that is correlated with the state: if the state is s_i , then the signal equals y_i with probability $p \in [0, 1]$ and equals y_0 with probability $1 - p$. Note that with probability p the person learns the state for sure, and with probability $1 - p$ the person learns nothing.

You might find this notation useful.

$$v(a, \pi) = \max_{b \in C(a)} \{u_2(b, s_1)(1 - \pi) + u_2(b, s_2)\pi\}, \quad (2)$$

and

$$E[u_1(a_i, s)] = \frac{1}{2}u_1(a_i, s_1) + \frac{1}{2}u_1(a_i, s_2). \quad (3)$$

- (a) Explain in words what $v(a, \pi)$ is in (2). If $a = a_1$, then what are the solutions to problem (2) at $\pi = 0$, $\pi = 1/2$, and $\pi = 1$?
- (b) Let $U(a_i, p)$ equal the two-period expected utility if the person chooses a_i and date 1, and then chooses the best available action at date 2, given a_i and what the person learns between the dates. Write down an expression for $U(a_i, p)$.

Consider this implication: “for any $1 > p_1 > p_0 > 0$,

$$U(a_1, p_0) \geq U(a_2, p_0) \Rightarrow U(a_1, p_1) > U(a_2, p_1).” \quad (4)$$

- (c) Explain in words what (4) asserts. What if anything does it imply about how the person's choice of a at date 1 varies with p ? Explain.
- (d) Now to the substance of the problem: Prove that the implication (4) holds.

3. A monopolist (principal) has a unit of a product with observable quality $0 < q < 1$, interpreted as the probability that the product will work (it breaks down with probability $1 - q$), and faces a buyer (agent) with evaluation θ in dollars for a functioning product. The evaluation $\theta \in \{\theta_\ell, \theta_h\}$, $0 < \theta_\ell < \theta_h$, is the buyer's private information, and $\mathbb{P}[\theta = \theta_h] = \beta \in (0, 1)$. The monopolist offers a menu of contracts, where each contract consists of a price p and a warranty w (a reimbursement if the product breaks down). The principal's utility for (p, w) is $p - (1 - q)w$ and the agent's when his type is θ is $qu(\theta - p) + (1 - q)u(w - p)$, $u' > 0$, $u'' < 0$. The agent's reservation utility is $u(0)$.

- (a) Using the revelation principle, set up the principal's problem of finding a menu (mechanism) that maximizes her expected profits. Explain how you use the revelation principle.
- (b) Assume from now on that $u(c) = -e^{-rc}$, $r > 0$ (CARA utility function). Derive the certainty equivalent for (p, w) and show that, without loss of generality, we can represent the buyer's preferences for (p, w) when his type is θ by $v(\theta, w) - p$, where $v(\theta, w) = -\frac{1}{r} \log(qe^{-r\theta} + (1 - q)e^{-rw})$. Show that v is strictly supermodular in (θ, w) .
- (c) Show that (i) any feasible menu entails a higher (p, w) for the high type than for the low type, and (ii) the participation constraint of the low type and the incentive constraint of the high type are binding at the optimum. Accompany your arguments with pictures.
- (d) Assume (b)–(c) and solve for the profit-maximizing menu. Discuss its properties.
- (e) Now assume that, in addition to p and w , the principal also chooses $q \in [0, 1]$ at a cost $c(q)$, where $c : [0, 1] \rightarrow \mathbb{R}_+$ is strictly increasing and convex with $c(0) = 0$. Without solving, set up the principal's problem for a general u (not necessarily CARA). Do you think (q, p, w) is higher for θ_h than for θ_ℓ at the optimum?

4. Consider a Cournot duopoly game with inverse demand given by

$$P(Q) = \max\{A - Q, 0\},$$

where Q is aggregate output. Both firms have *constant-returns* technologies: firm 2's unit cost is k_2 ; firm 1's unit cost can take on one of two *equally-likely* values, k_1^L or k_1^H . *Firm 1 knows the cost of both firms, but firm 2 only knows its own cost.* Assume that

- (i) $0 < k_1^L < k_2 < k_1^H$,
- (ii) $\frac{1}{2}k_1^H + \frac{1}{2}k_1^L = k_2$, and
- (iii) $A > 2k_1^H$ and $A > 2k_2$.

Assumption (ii) asserts that the expected unit cost of firm 1 equals firm 2's commonly-known unit cost. Assumption (iii) implies that firm 2 and both types of firm 1 produce positive output in a Bayesian equilibrium of the game.

- (a) What is a strategy for firm 1 in this Bayesian game? What is a strategy for firm 2? Write down the definition of a Bayesian equilibrium for *this* game.
- (b) Identify the best reply function for each firm.
- (c) Identify a Bayesian equilibrium for this game. (Suggestion: keep assumption (ii) in mind.)

Now suppose that, after firm 1 learns its cost, but before the firms simultaneously choose outputs, firm 1 must send a message to firm 2 about its unit cost. A low-cost firm 1 can send message L (its unit cost is low) or LH (its cost is either high or low); and a high-cost firm 1 can send message H (its unit cost is high) or LH (its cost is either high or low). In other words, firm 1 can't lie about its cost, but doesn't have to tell the whole truth. Firm 2 observes firm 1's message before the firms simultaneously choose outputs.

- (d) What is an *assessment* in this Bayesian game? What is a strategy for firm 1? What is a strategy for firm 2?
- (e) Is there a weak perfect Bayesian equilibrium (wpBE) in which both types of firm 1 send the message LH ? Explain. Is there a wpBE in which firm 2 fails to learn firm 1's cost? (It is possible, *but not recommended*, to solve fully for the set of wpBE; the best answers simply use the definition of a wpBE, results from parts (a)-(d), and your knowledge of optimization.)